

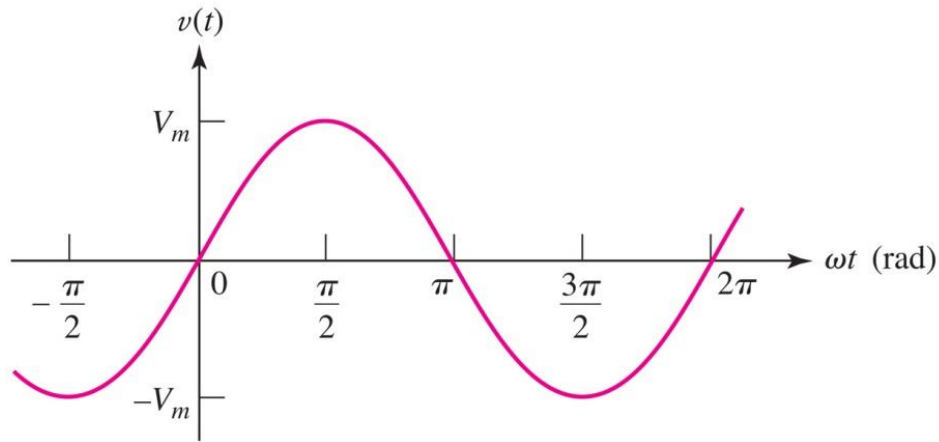
William H. Hayt, Jr. • Jack E. Kemmerly • Steven M. Durbin



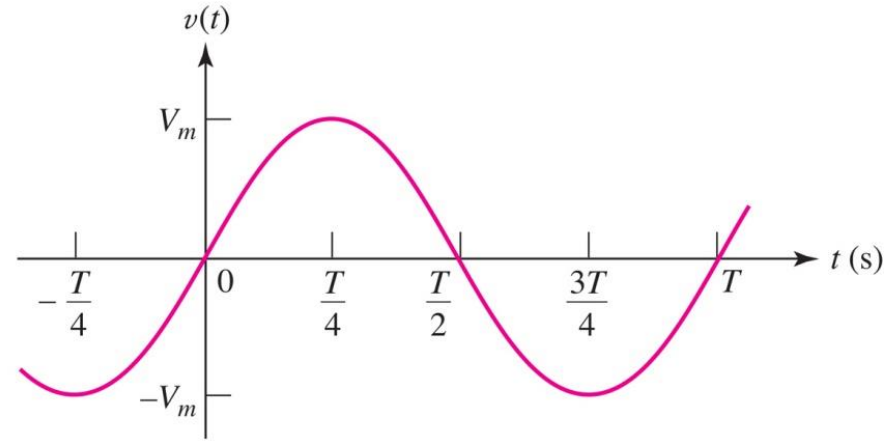
Engineering Circuit Analysis

Eighth Edition

AC circuits: **sinusoidal steady state analysis**



(a)



(b)

$$v(t) = V_m \sin(\omega t + \theta)$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$f = 1/T$$

A wave, which moves up and down with the dependence of **sine** or **cosine** is called **harmonic**

Sine and Cosine Addition and Subtraction Formulas

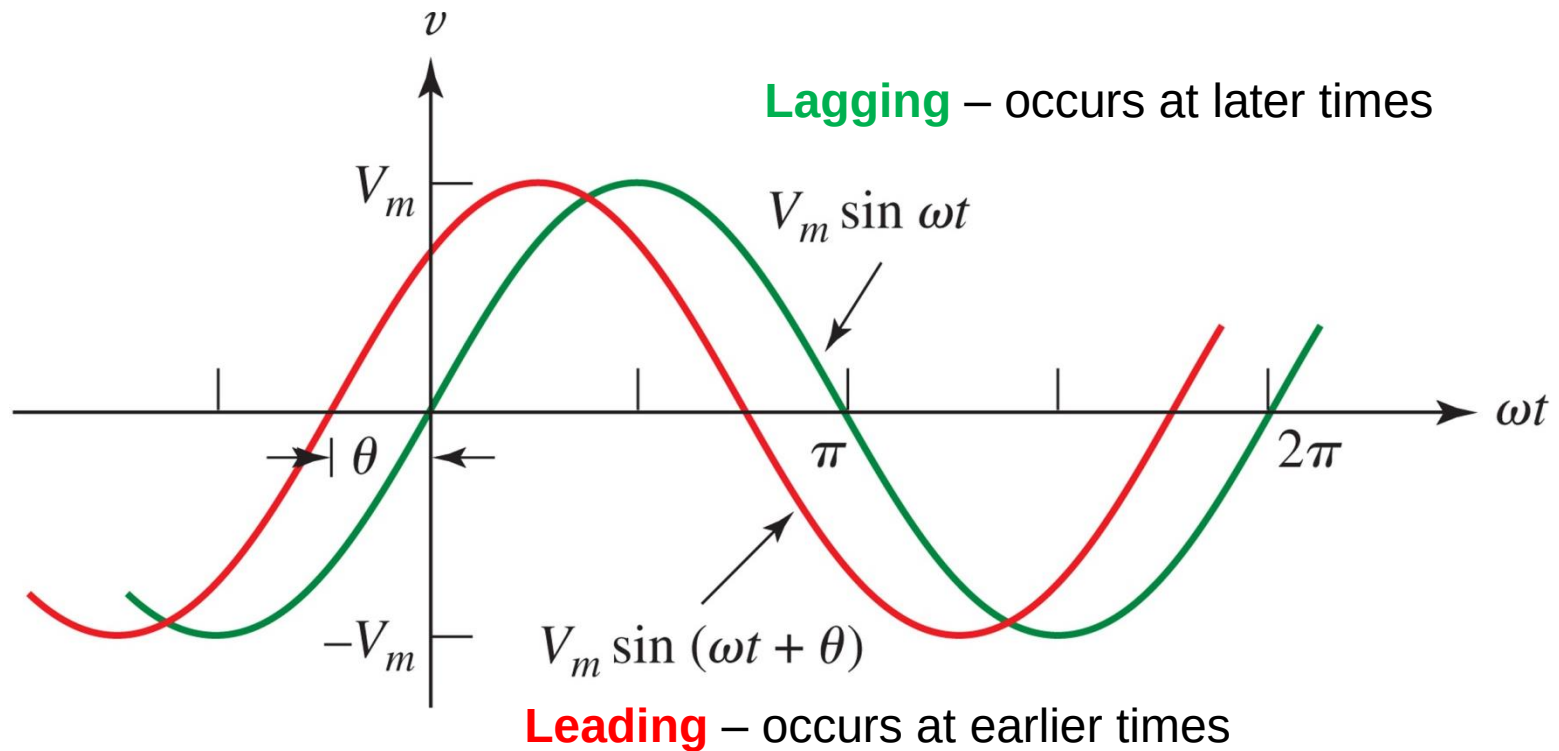
$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

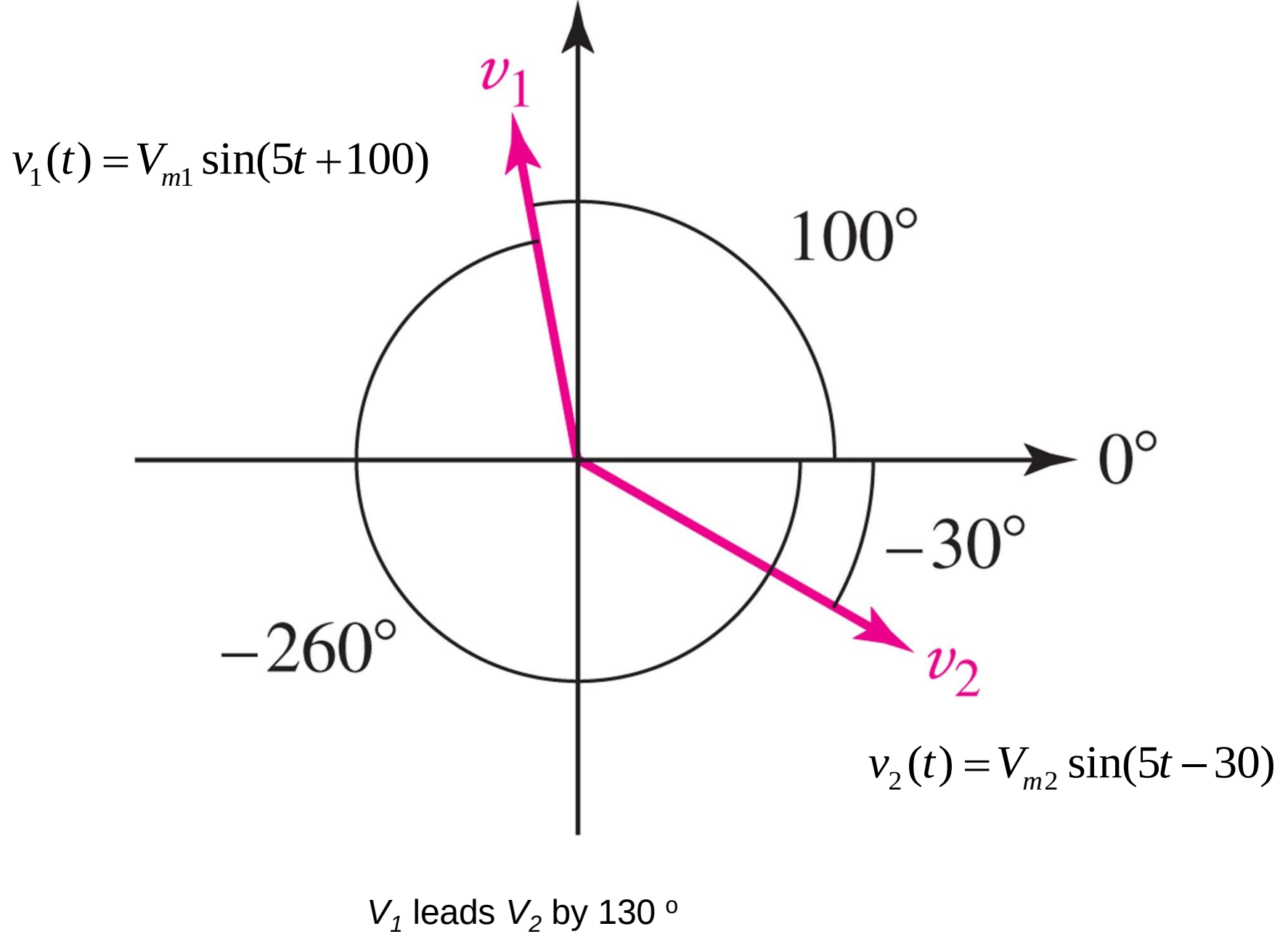
$$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

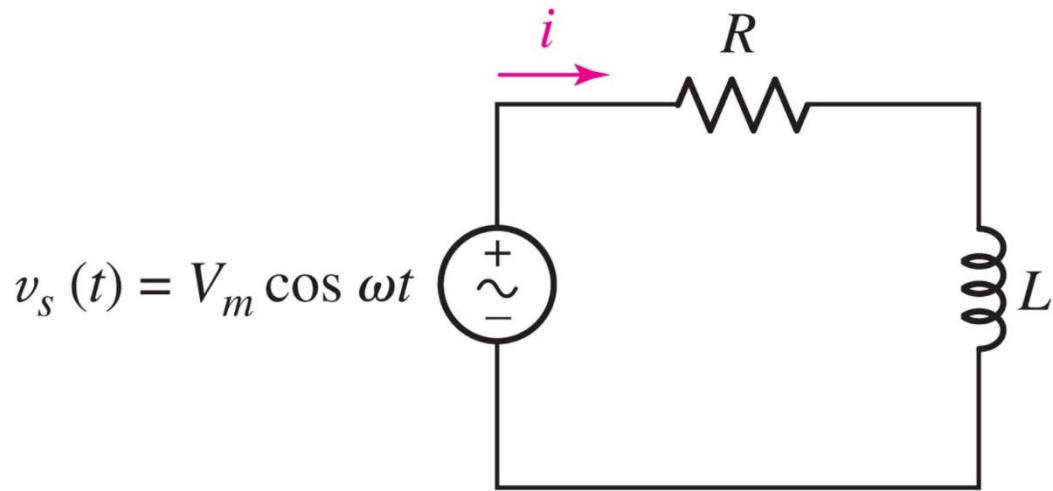
$$v(t) = V_m \sin(\omega t \pm \theta)$$





Steady State Response $\equiv$ Forced Response

Consider a circuit below after all transients died out



$$L \frac{di}{dt} + Ri - V_m \cos \omega t = 0$$



$$i(t) = I_1 \cos \omega t + I_2 \sin \omega t$$



Alternatively

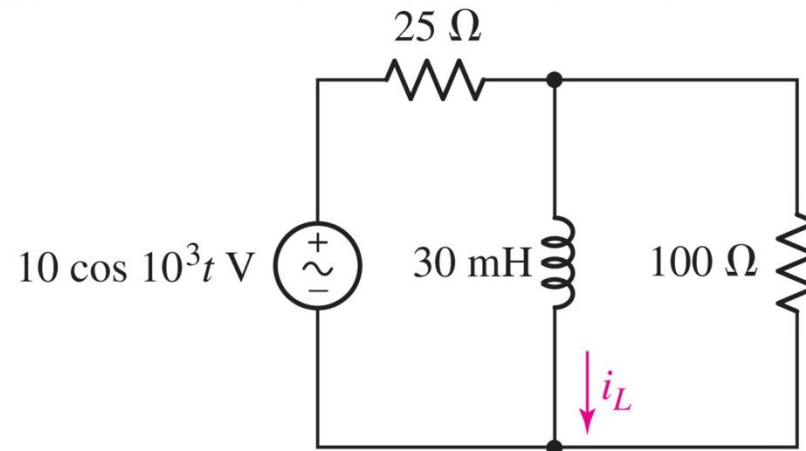
$$i(t) = A \cos(\omega t + \theta)$$



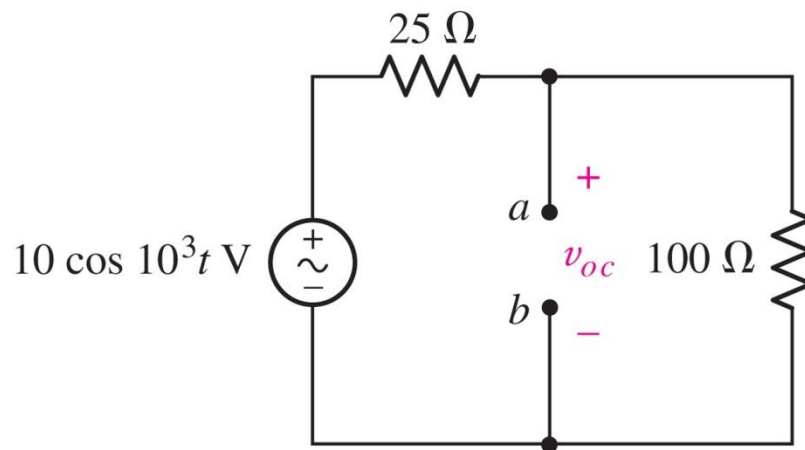
$$i(t) = \frac{RV_m}{R^2 + \omega^2 L^2} \cos \omega t + \frac{\omega L V_m}{R^2 + \omega^2 L^2} \sin \omega t$$

$$i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos\left(\omega t - \tan^{-1} \frac{\omega L}{R}\right)$$

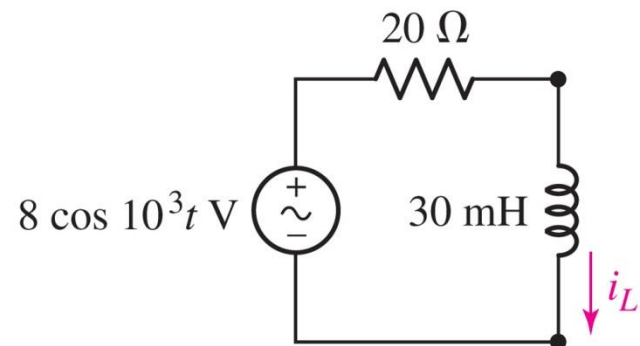
Find i_L in a circuit below after all transients died out



(a)



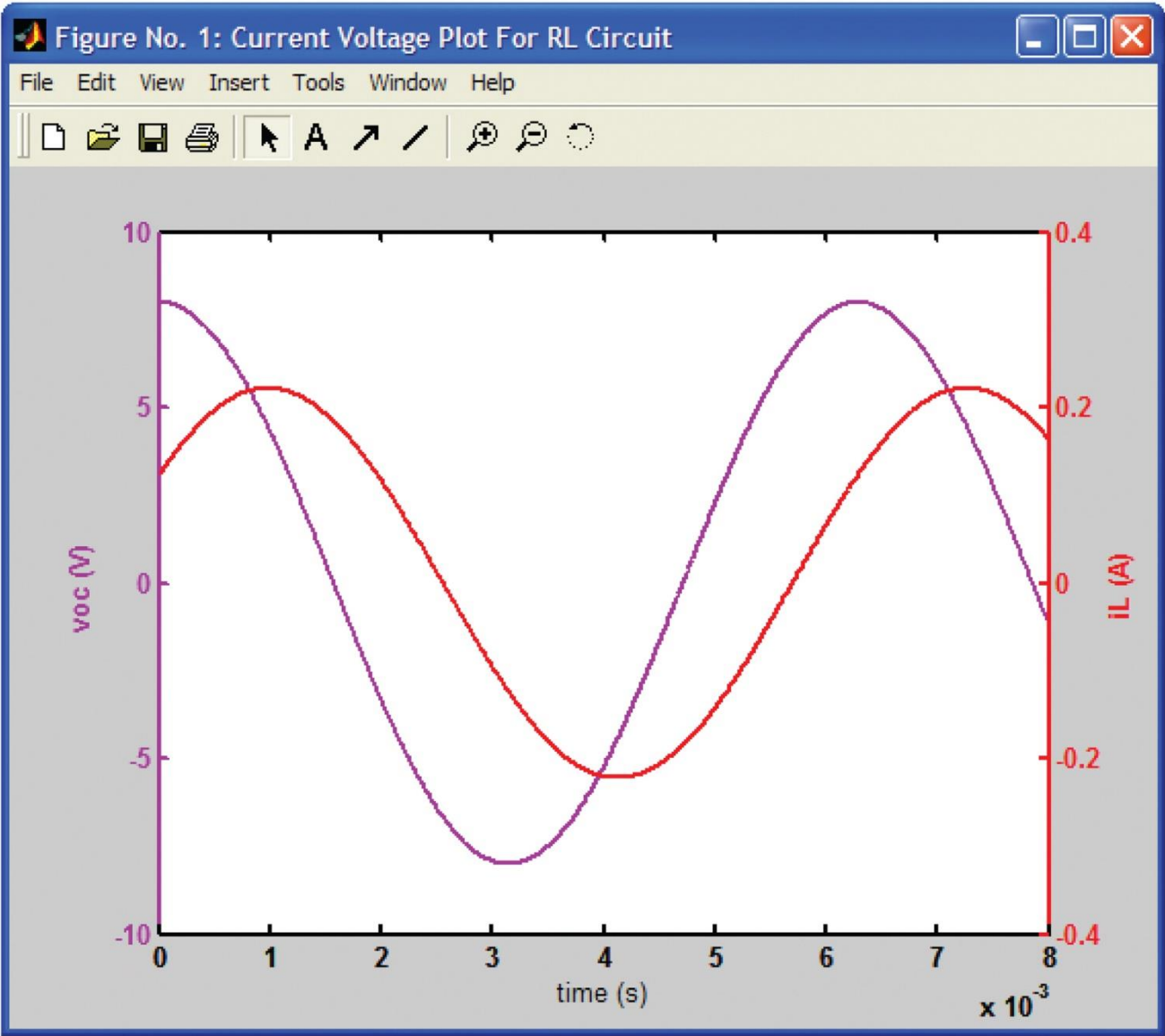
(b)



(c)

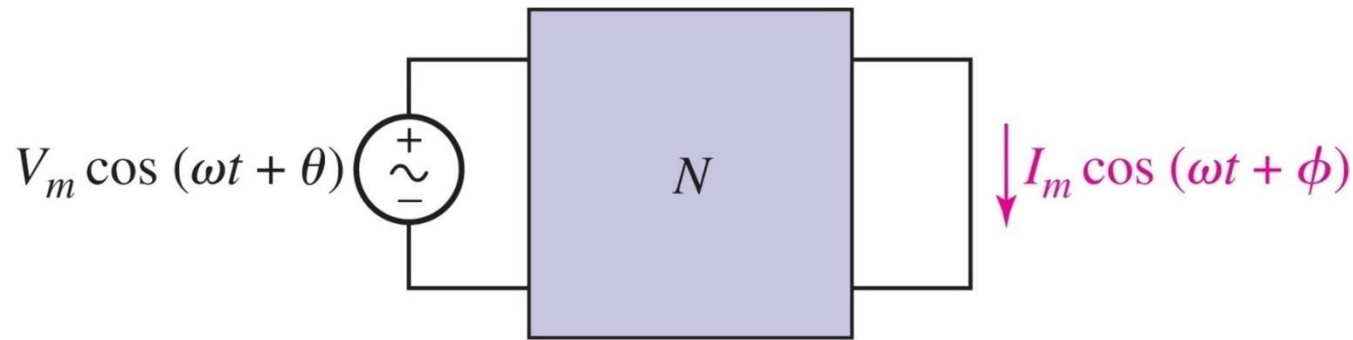
Fig. 1006

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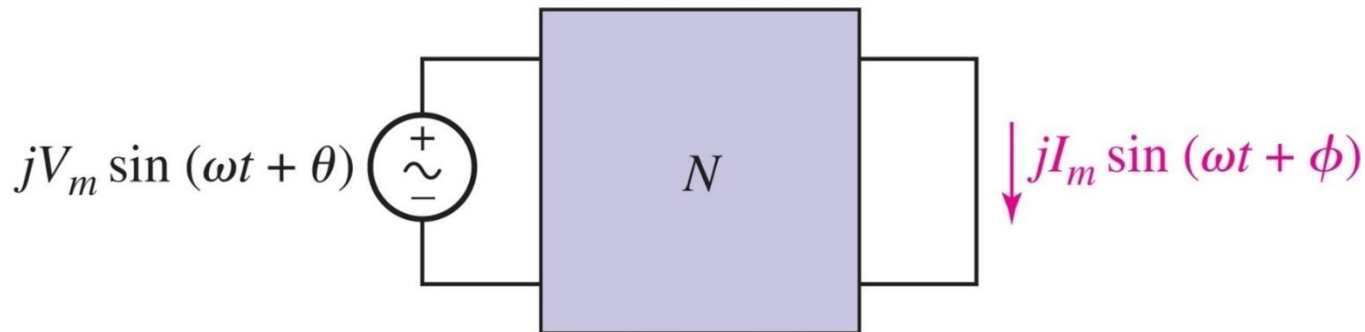


Complex forcing function

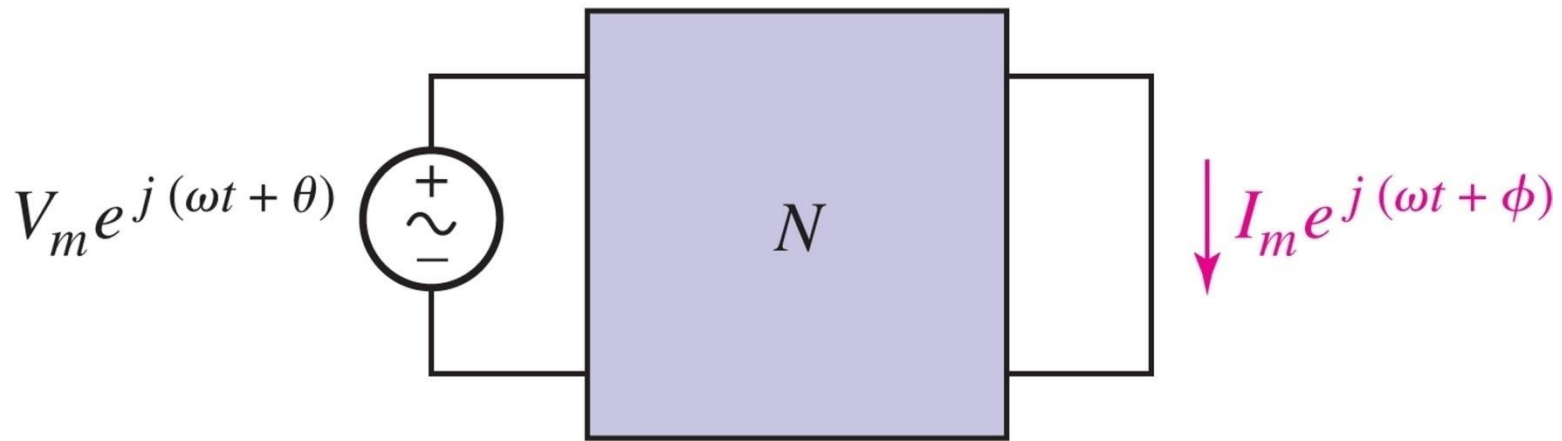
Passive linear network



A sinusoidal forcing function always produces a sinusoidal forced response of the same frequency in a linear circuit

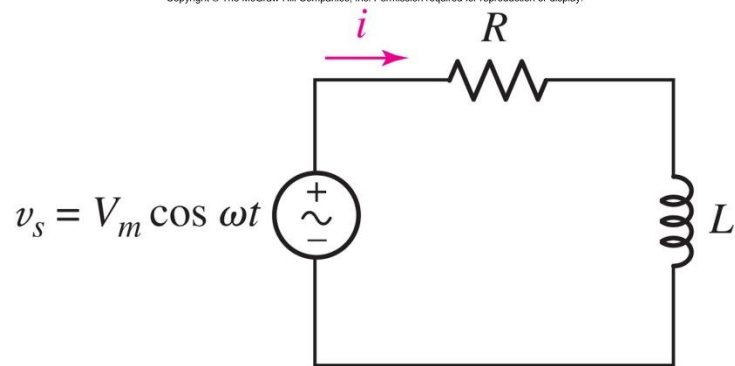


$$V_m \cos(\omega t + \theta) + jV_m \sin(\omega t + \theta) \implies I_m \cos(\omega t + \phi) + jI_m \sin(\omega t + \phi)$$



We have substituted real forcing function with complex one to simplify further calculations

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$$L \frac{di}{dt} + Ri - v_s = 0$$

$$v_s = V_m e^{j\omega t}$$

$$i = I_m e^{j(\omega t + \phi)}$$

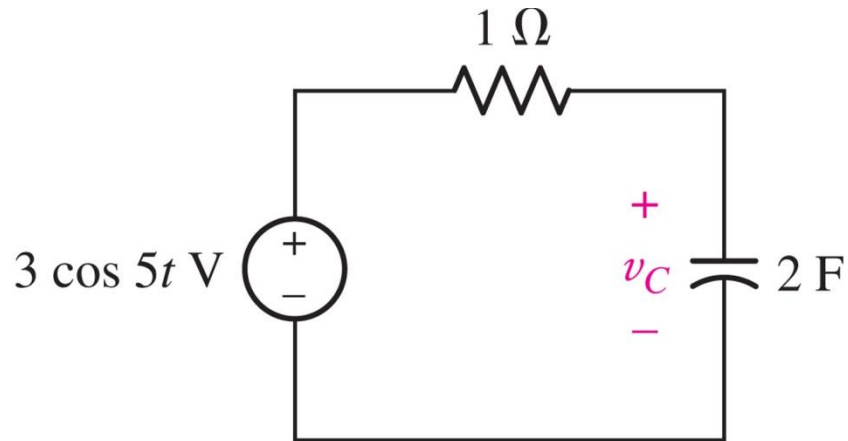
$$I_m e^{j\phi} = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} e^{j(-\tan^{-1} \frac{\omega L}{R})}$$



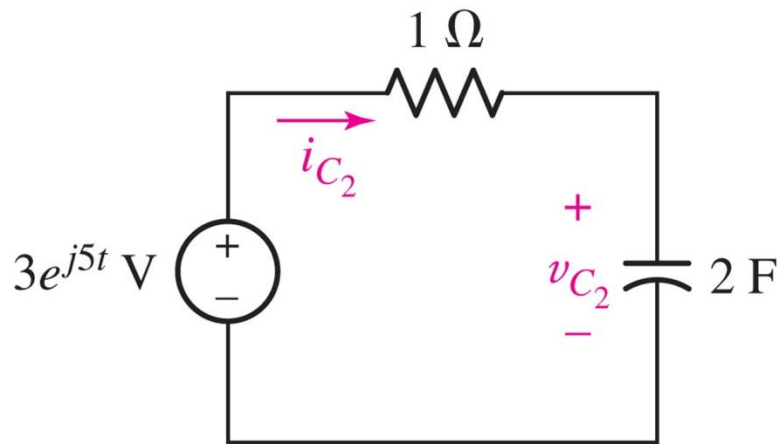
$$i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t - \tan^{-1} \frac{\omega L}{R})$$

$$I_m \angle \phi = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \angle -\tan^{-1}(\omega L / R)$$

Find steady-state capacitor voltage



(a)



(b)

$$3 \cos 5t \rightarrow 3e^{j5t}$$

$$-3e^{j5t} + 1 \cdot 2 \frac{dv_{C_2}}{dt} + v_{C_2} = 0$$

$$v_{C_2} = V_m e^{j(5t+\phi)}$$

$$V_m e^{j\phi} = 0.299 e^{-j \tan^{-1} 10}$$

$$v_{C_2} = 0.299 \cos(5t - 84.3^\circ)$$

$$0.299 \angle -84.3^\circ$$

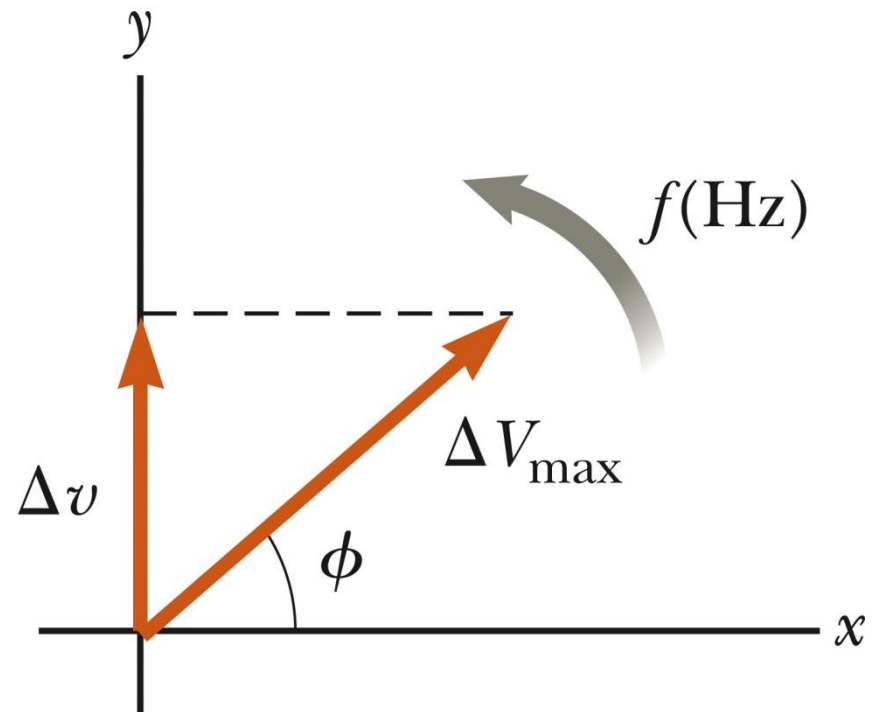
- To account for the different phases phasor diagram is used.
- **Phasor** is a special vector that represents a wave.

Length: The length of each phasor represents the amplitude of the alternating quantity: V_m for the voltage and I_m for the current.

Projection: The projection of each phasor on the axis represents the real (x) and imaginary (y) values of the alternating quantity at time t : v for the voltage and i for the current.

Rotation angle: The rotation angle of each phasor is equal to the phase of the alternating quantity at time t .

Angular speed: Both current and potential difference phasors rotate counterclockwise about the origin with an angular speed equal to the angular frequency ω of v and i .



Phasor transformation from time domain to frequency domain

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$$i(t) = I_m \cos(\omega t + \phi)$$

← Time-domain representation

$$i(t) = \operatorname{Re}\{I_m e^{j(\omega t + \phi)}\}$$

$$\mathbf{I} = I_m e^{j\phi}$$

← Frequency-domain representation

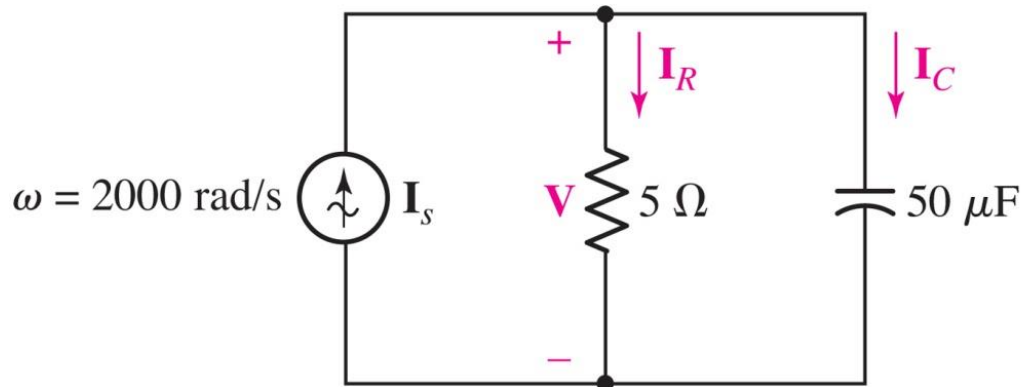
$$\mathbf{I} = I_m \underline{\angle \phi}$$

← Phasor

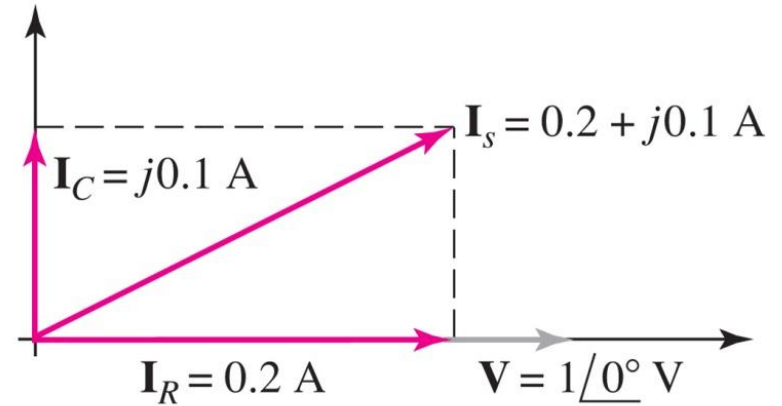
Let $\omega=2000$ rad/s and $t=1$ ms. Find the instantaneous value of each of the currents given here in phasor form: (a) $j10$ A; (b) $20+j10$ A; (c) $20+j10\angle 20^\circ$

Phasor Diagrams

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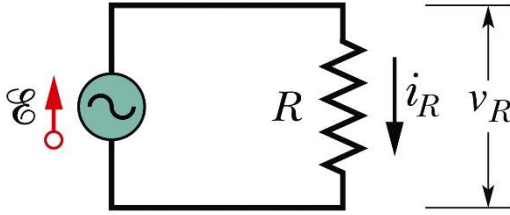


(a)



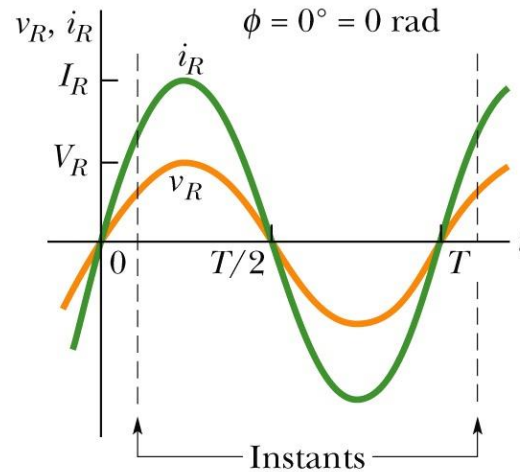
(b)

Resistive Load



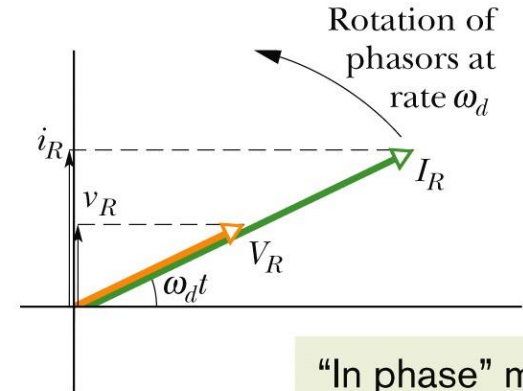
A resistor is connected across an alternating-current generator.

For a resistive load, the current and potential difference are in phase.



Instantaneous values represented in (b)

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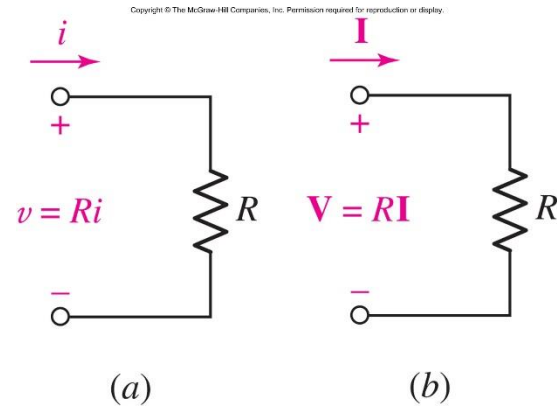
“In phase” means that they peak at the same time.

$$V_R = I_R R \quad (\text{resistor}).$$

$$v_R = V_m \sin \omega_d t$$

$$i_R = \frac{v_R}{R} = \frac{V_m}{R} \sin \omega_d t$$

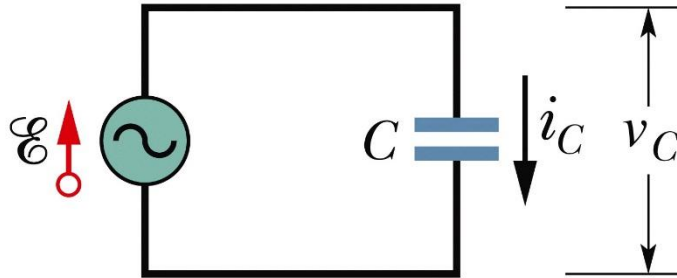
$$\mathbf{V} = R\mathbf{I}$$



(a) The current i_R and the potential difference v_R across the resistor are plotted on the same graph, both versus time t . They are in phase and complete one cycle in one period T . (b) A phasor diagram shows the same thing as (a).

Capacitive Load

A capacitor is connected across an alternating-current generator.



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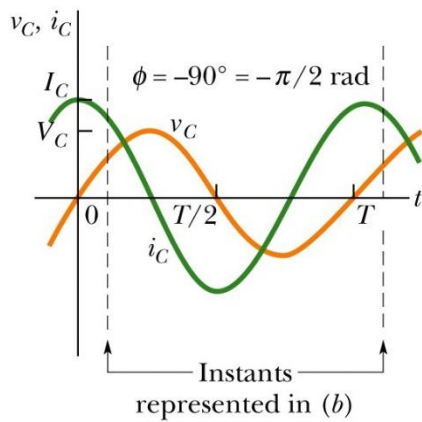
$$\varepsilon = v_C = V_m e^{j\omega t}$$

$$i_C = C \frac{dv_C}{dt} = V_m j\omega C e^{j\omega t} = j\omega C v_C$$

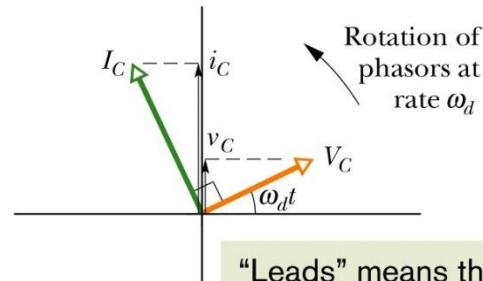
$$\mathbf{I} = j\omega C \mathbf{V}$$

$$\phi = \pi / 2$$

For a capacitive load, the current leads the potential difference by 90° .



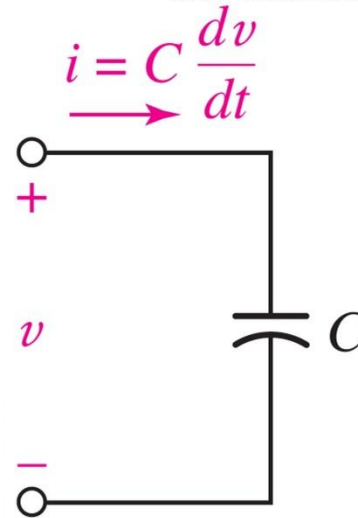
(a)



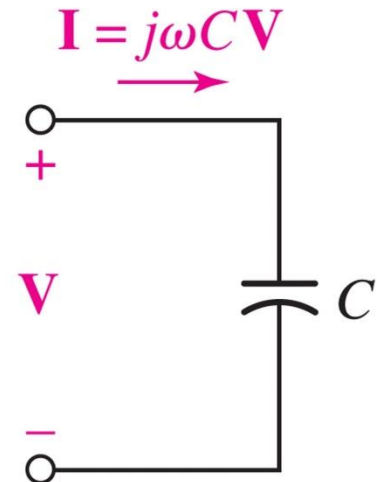
"Leads" means that the current peaks at an *earlier* time than the potential difference.

(b)

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(a)



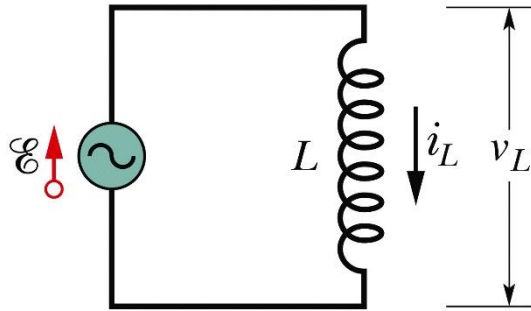
(b)

(a) The current in the capacitor leads the voltage by 90° ($= \pi/2$ rad).

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Inductive Load

An inductor is connected across an alternating-current generator.



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For an inductive load, the current lags the potential difference by 90° .

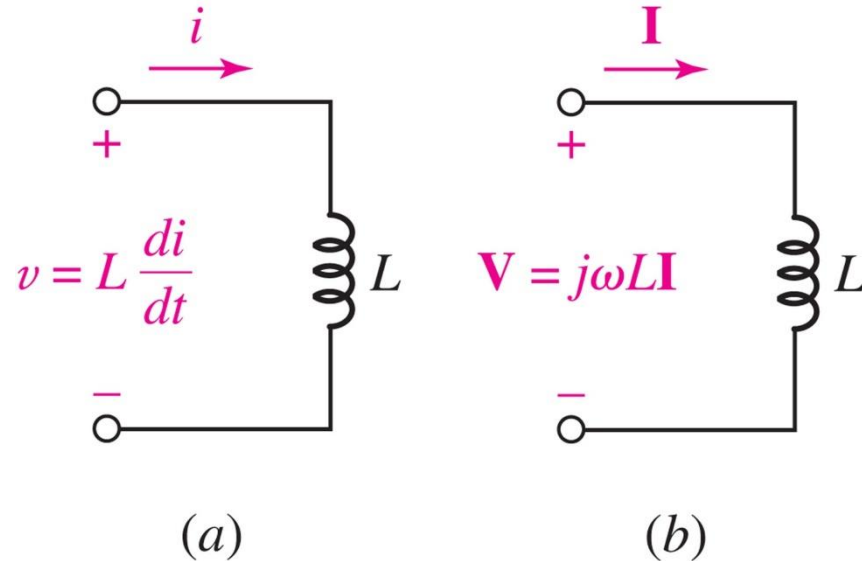
$$v_L = V_m e^{j(\omega t + \theta)} \quad v_L = L \frac{di_L}{dt} = L \frac{d}{dt} (I_m e^{j(\omega t + \phi)})$$

$$V_m e^{j\theta} = j\omega L I_m e^{j\phi}$$

$$\mathbf{V} = j\omega L \mathbf{I}$$

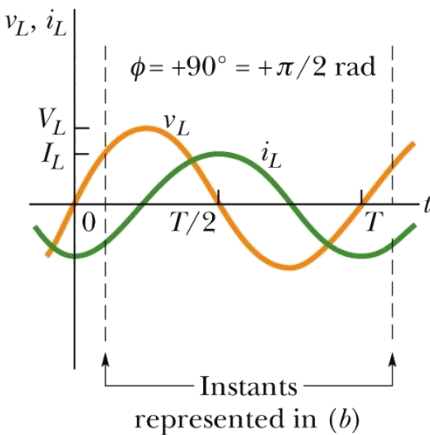
$$\phi = \pi / 2$$

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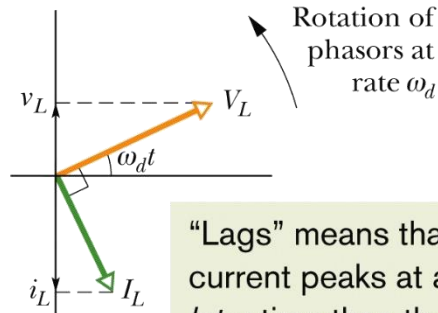


(a)

(b)



(a)



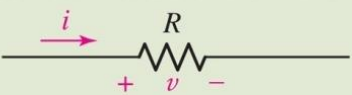
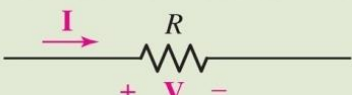
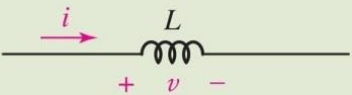
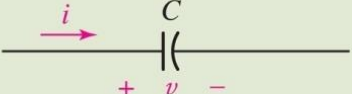
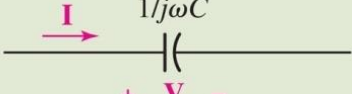
(b)

“Lags” means that the current peaks at a *later* time than the potential difference.

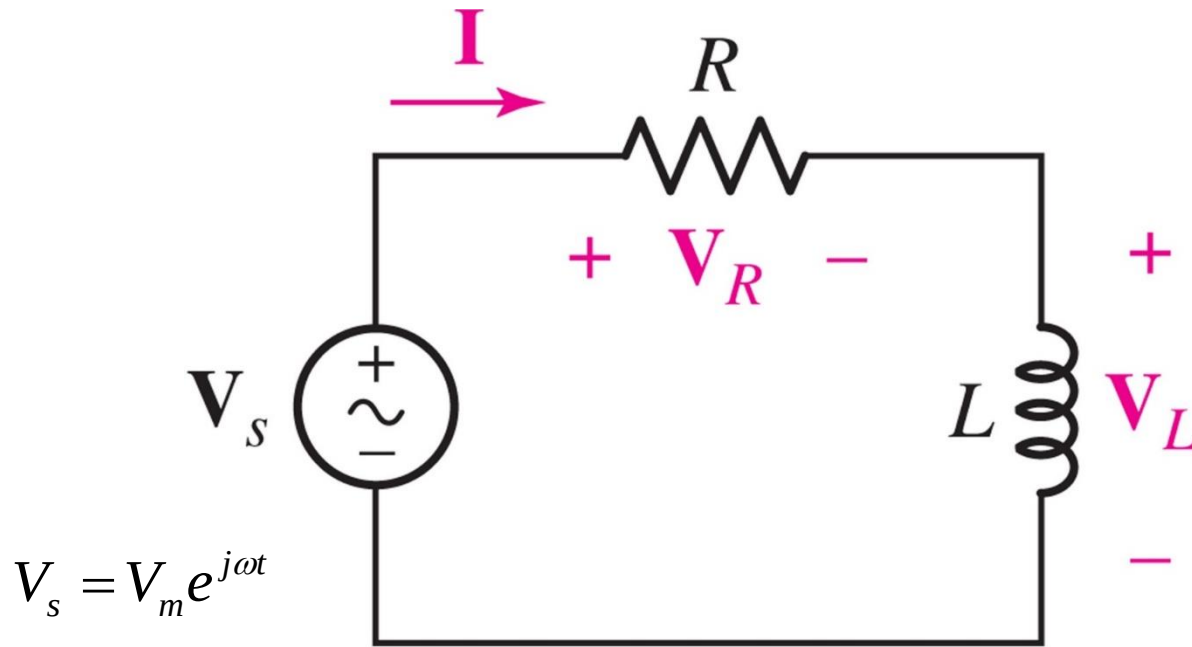
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(a) The current in the inductor lags the voltage by 90° ($= \pi/2$ rad).

TABLE 10.1 Comparison of Time-Domain and Frequency-Domain Voltage-Current Expressions

Time Domain		Frequency Domain	
	$v = Ri$	$\mathbf{V} = R\mathbf{I}$	
	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$	
	$v = \frac{1}{C} \int i \, dt$	$\mathbf{V} = \frac{1}{j\omega C}\mathbf{I}$	

Passive sign convention and Kirchhoff's laws apply to phasors as well

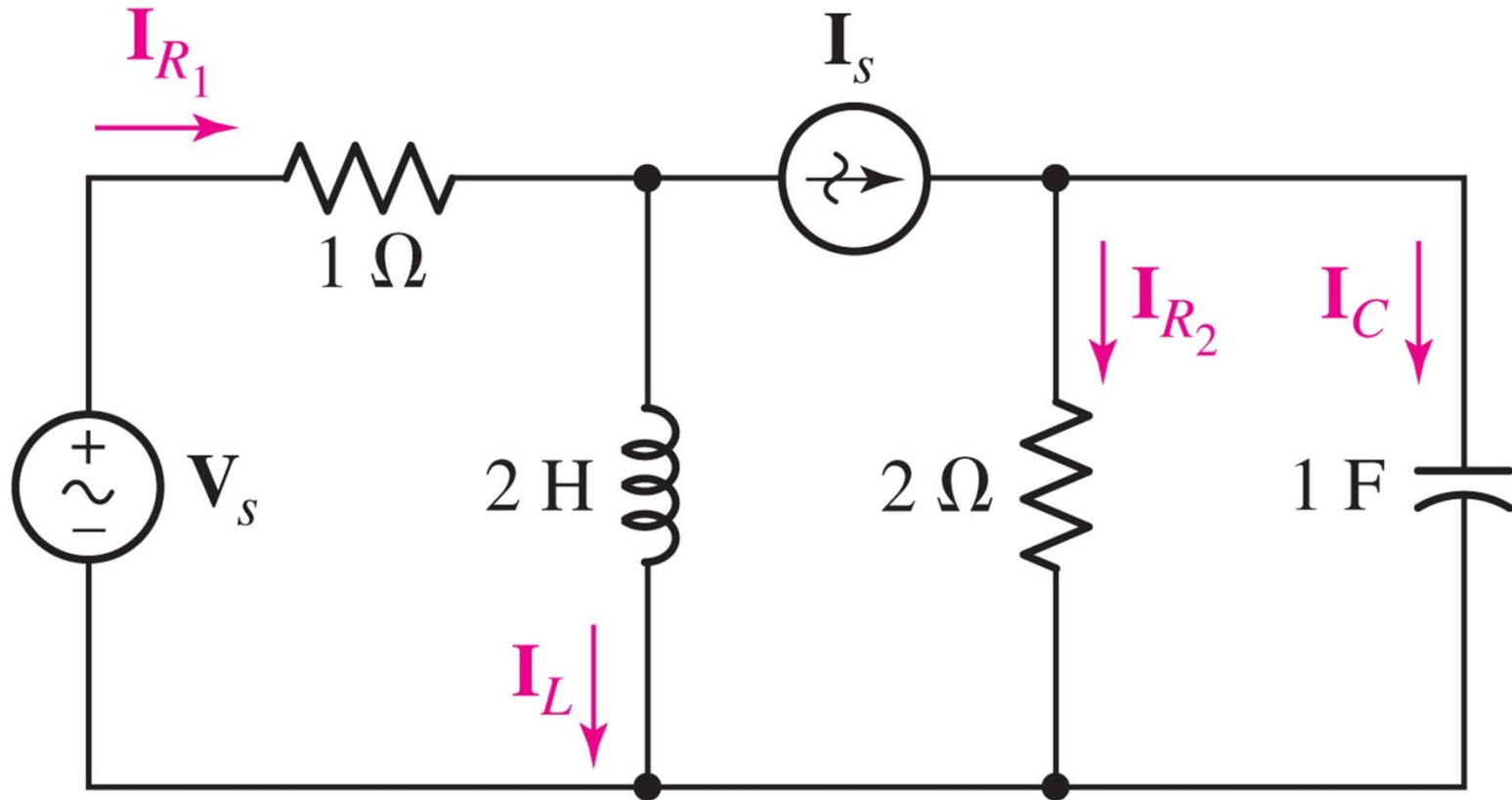


$$-V_s + V_R + V_L = 0$$

$$IR + j\omega L I = V_s$$

$$I = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} e^{-\tan^{-1}(\frac{\omega L}{R})}$$

Both sources operate at frequency $\omega=1$ rad/s. If $\mathbf{I}_C=2\angle 28^\circ$ and $\mathbf{I}_L=3\angle 53^\circ$, find $\mathbf{I}_s$, $\mathbf{V}_s$ and $i_{R1}(t)$.



$$3\angle -62^\circ\text{ A}; 3.71\angle -4.5^\circ\text{ V}; 3.22\cos(t-4.5^\circ)\text{ A}$$

Impedance and Admittance

Resistor

$$\mathbf{V} = R\mathbf{I}$$

Inductor

$$\mathbf{V} = j\omega L\mathbf{I}$$

Capacitor

$$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

Impedance ($\mathbf{Z}$) – ratio of complex voltage to complex current

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

$$\mathbf{Z} = \underset{\substack{\nearrow \\ \text{Resistance}}}{R} + j \underset{\substack{\nearrow \\ \text{Reactance}}}{X} = |\mathbf{Z}|e^{j\theta}$$

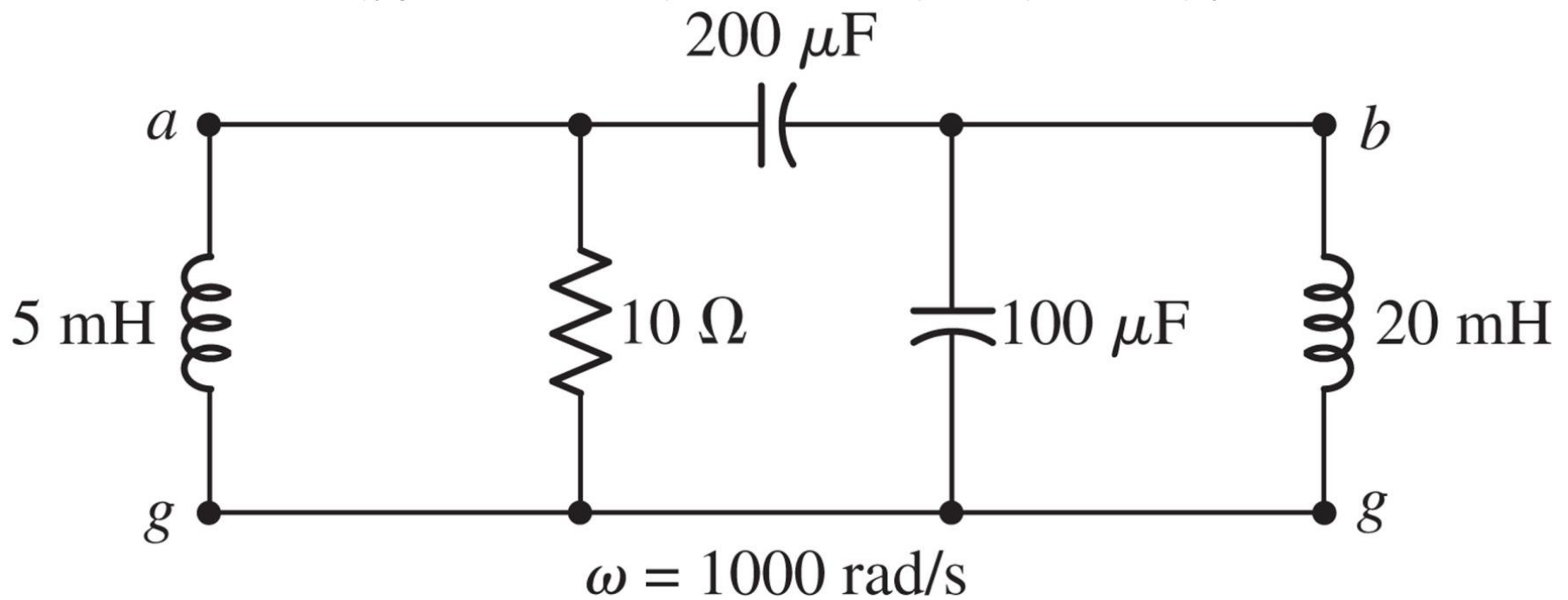
$\mathbf{Z}$ can be combined using series and parallel rules for resistances

Admittance ($\mathbf{Y}$) – ratio of complex current to complex voltage

$$\mathbf{Y} = \frac{\mathbf{I}}{\mathbf{V}}$$

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \underset{\substack{\nearrow \\ \text{Conductance}}}{G} + j \underset{\substack{\nearrow \\ \text{Susceptance}}}{B}$$

Find Z between a-g; b-g; a-b.



$$2.81+j4.49; 1.798-j1.124; 0.1124-j3.82$$

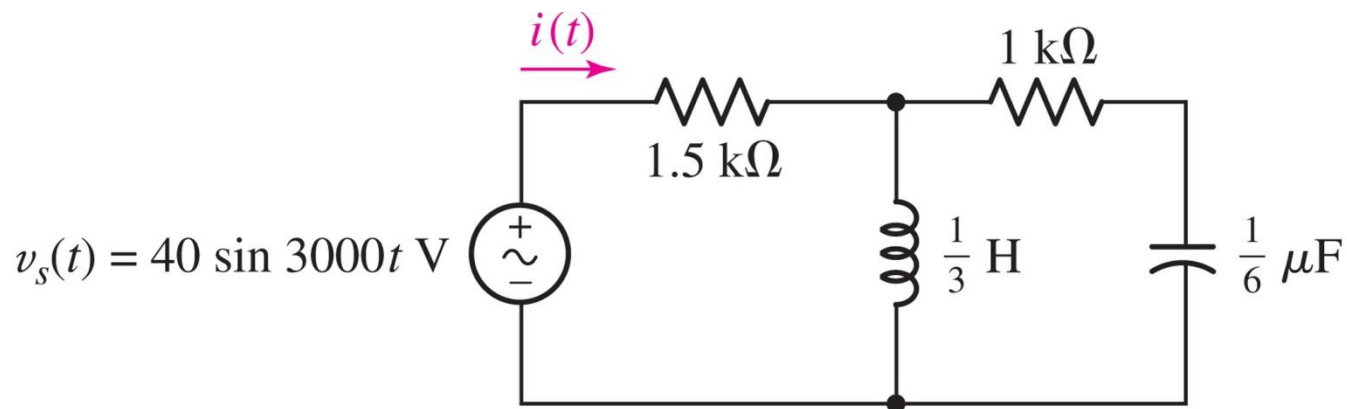
Find $i(t)$.

$$\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$$

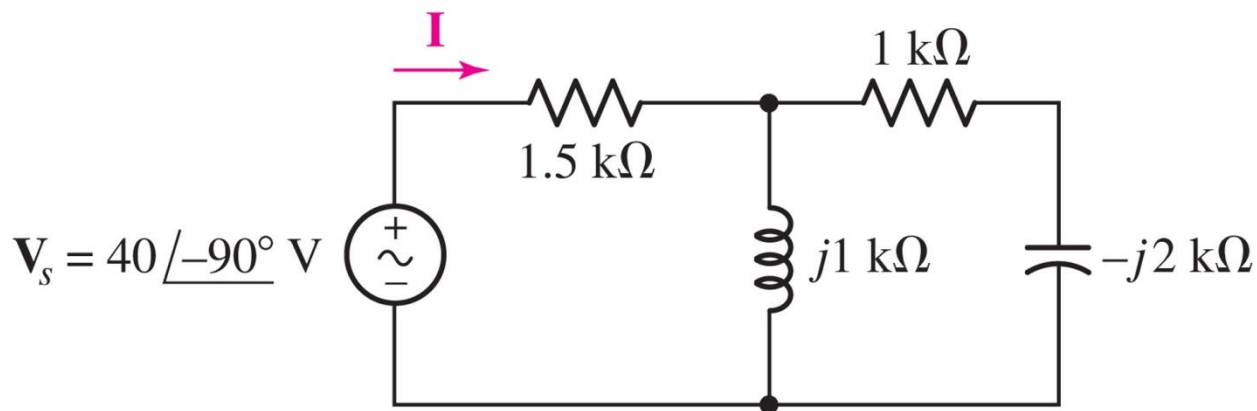
$$\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cot \theta = \tan\left(\frac{\pi}{2} - \theta\right)$$

$$\tan \theta = \cot\left(\frac{\pi}{2} - \theta\right)$$

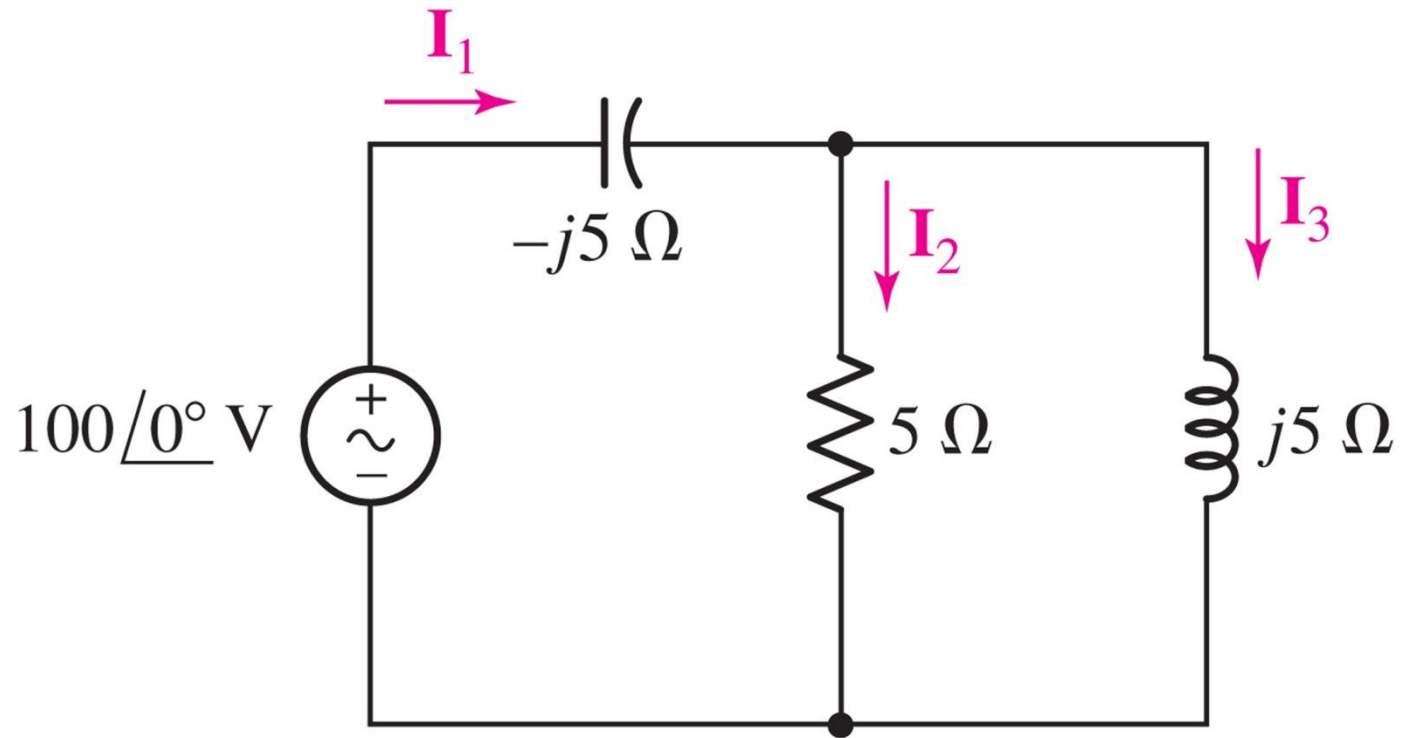


(a)



(b)

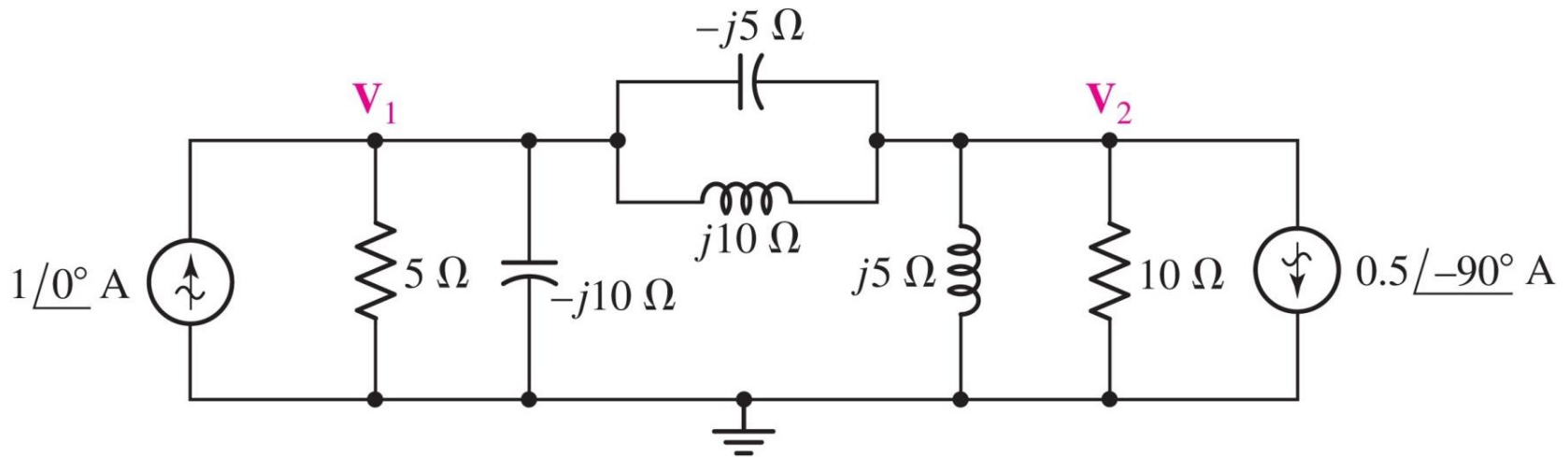
Find I_1 , I_2 , I_3 .



$$28.3 \angle 45^\circ; 20 \angle 90^\circ; 20 \angle 0^\circ$$

Nodal and Mesh analysis

Find $v_1(t)$ and $v_2(t)$.



$$\frac{V_1}{5} + \frac{V_1}{-j10} + \frac{V_1 - V_2}{-j5} + \frac{V_1 - V_2}{j10} = 1 + j \cdot 0$$

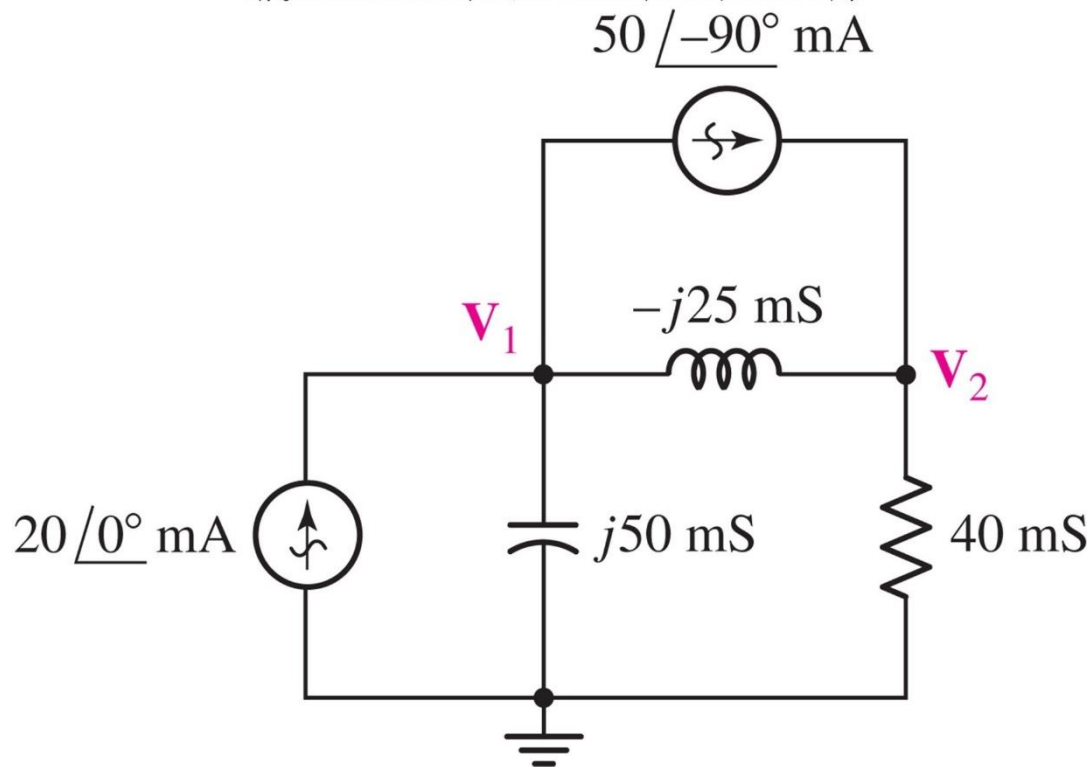
$$V_1 = 2.24e^{-j63.4}$$

$$\frac{V_1 - V_2}{-j5} + \frac{V_1 - V_2}{j10} = \frac{V_2}{j5} + \frac{V_2}{10} + (-0.5j)$$

$$V_2 = 4.47e^{j116.6}$$

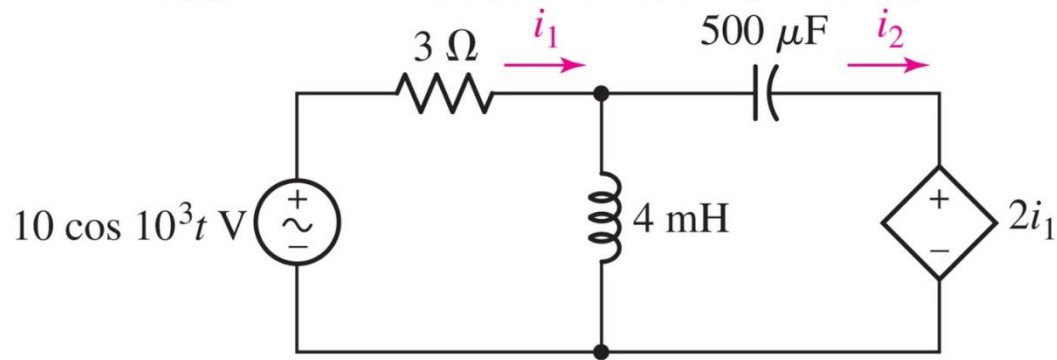
Nodal and Mesh analysis

Use nodal analysis to find $\mathbf{v}_1$ and $\mathbf{v}_2$ in frequency domain.

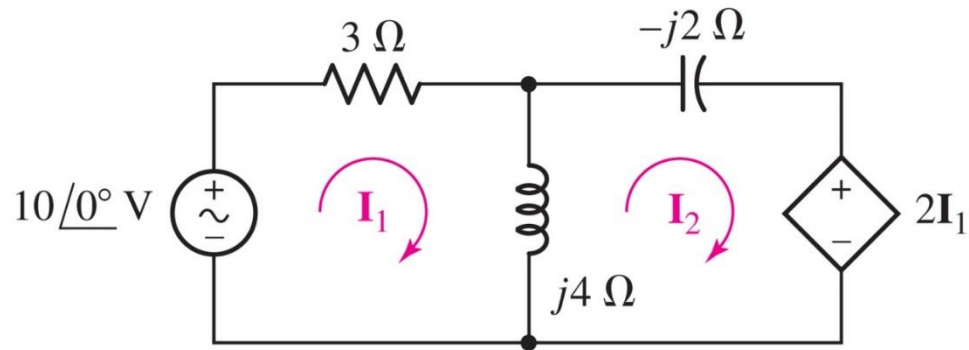


$$1.062 \angle 23.3^\circ \text{ V}; 1.593 \angle -50^\circ \text{ V}$$

Nodal and Mesh analysis



(a)

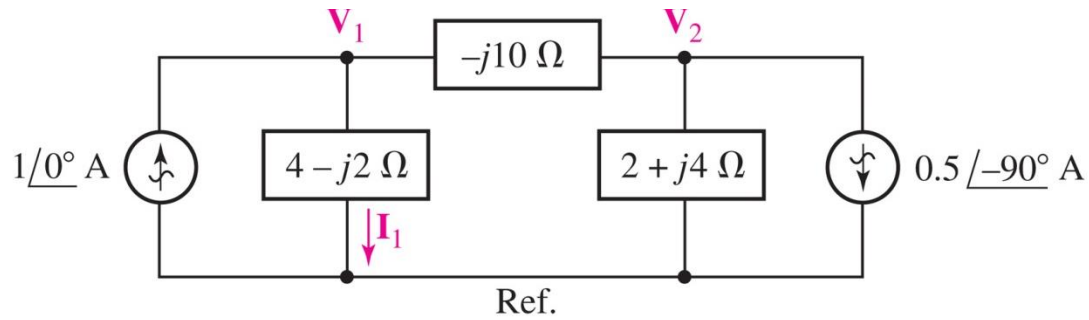


(b)

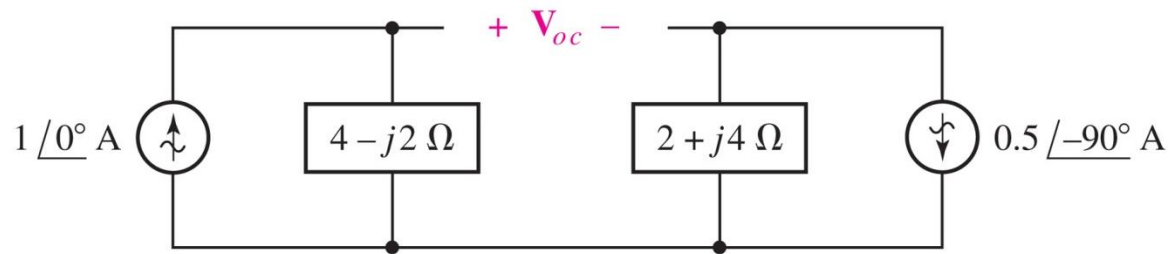
$$\mathbf{I}_1 = 1.24e^{j29.7}$$

$$\mathbf{I}_2 = 2.77e^{j56.3}$$

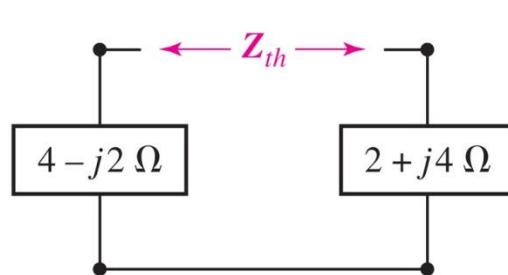
Norton's and Thevenin's theorems



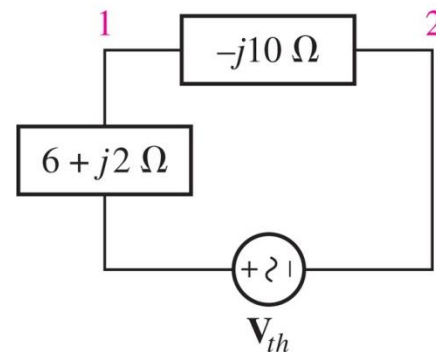
(a)



(b)



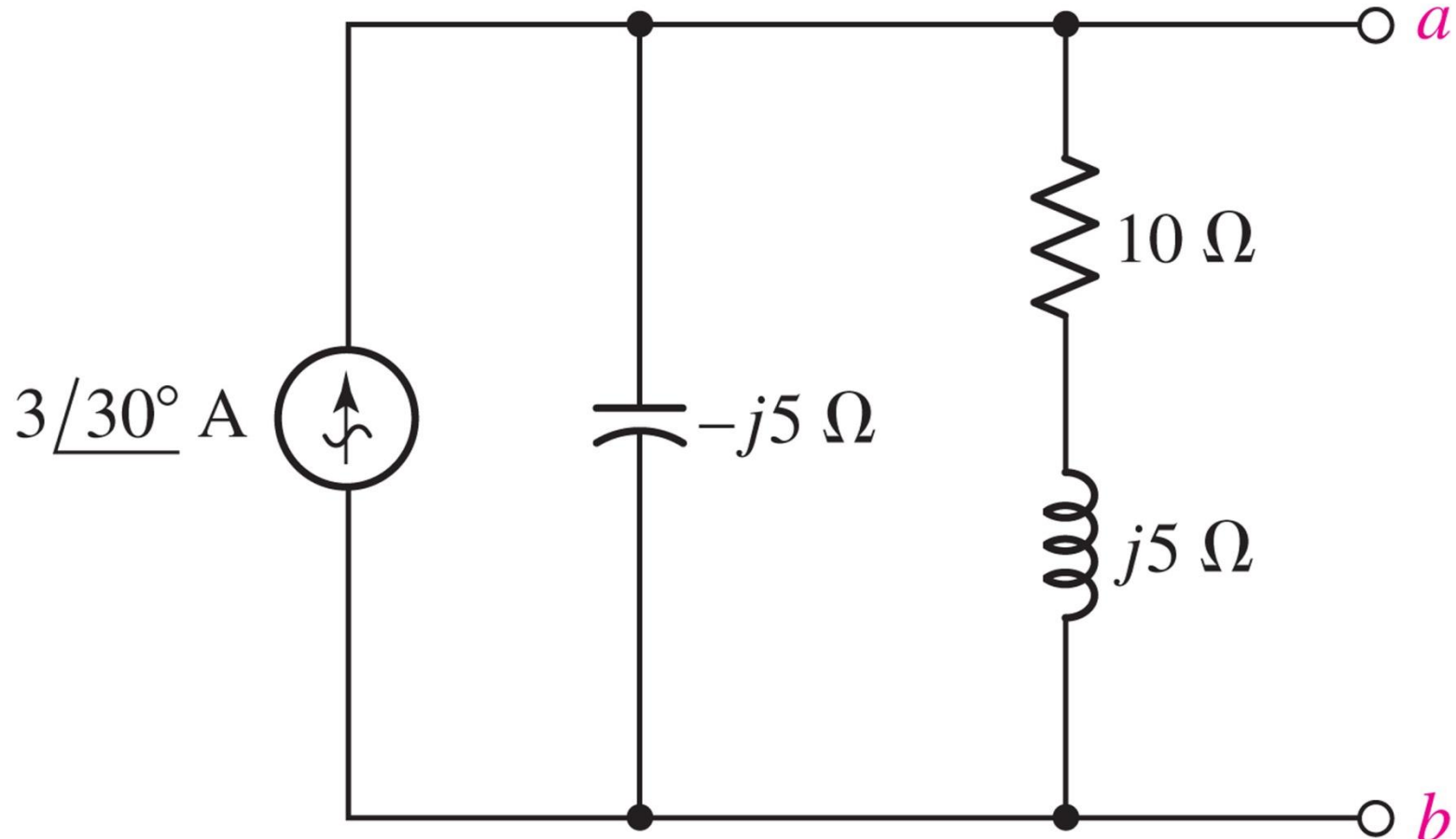
(c)



(d)

Find Thevenin's equivalent.

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Superposition principle

Find current i through 4 Ohms resistor.

